

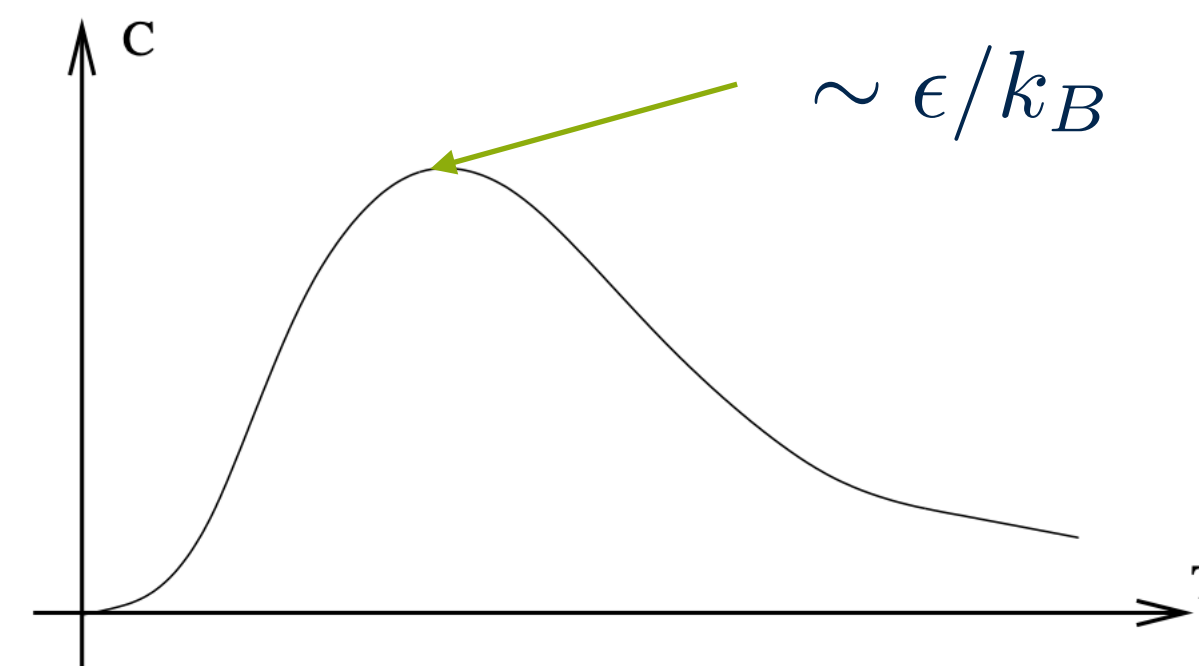
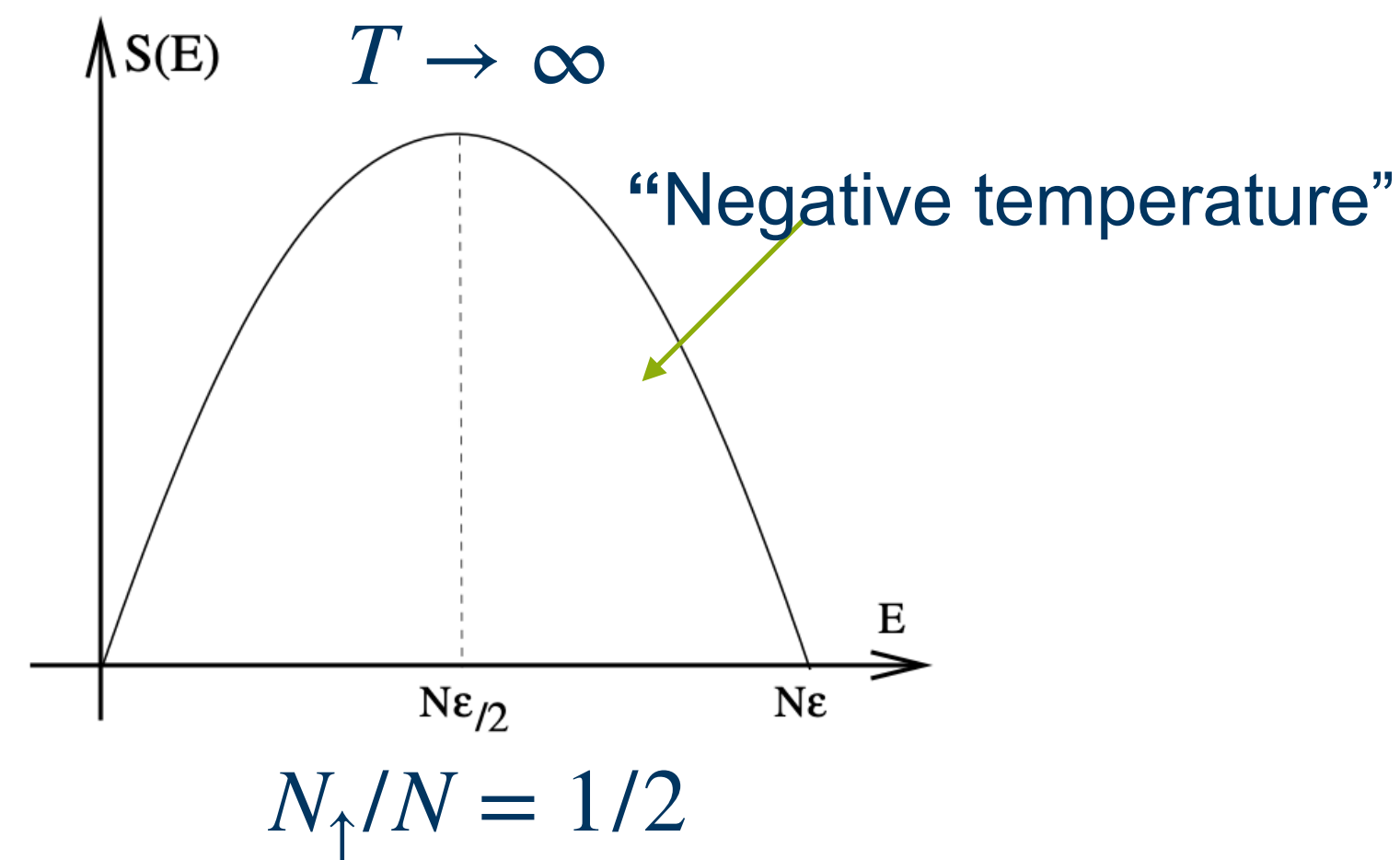
THE CANONICAL ENSEMBLE

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SUMMARY

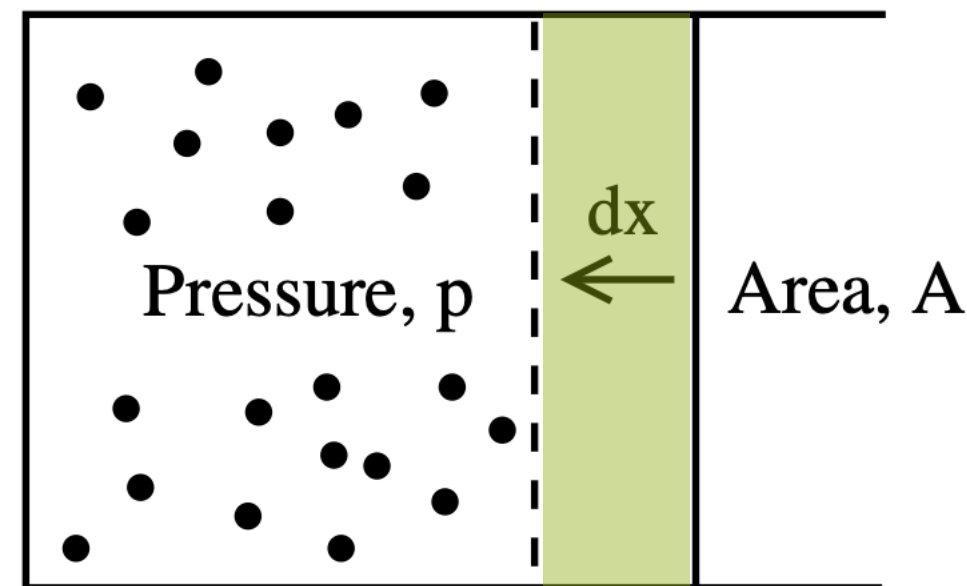
- $\frac{1}{T} = \frac{\partial S}{\partial E}$; with this definition, energy flows in the direction we expect (from hot to cold)
- $C = \frac{\partial E}{\partial T}$; heat capacity is a link between the micro (S) and the macro (E, T)-world; $C \propto N$, number of particles
- Two-state system: N non-interacting particles, either spin up or spin down; we calculated S(E) vs E and C vs T



Spin contribution to heat capacity; normally the contribution from phonons and conduction electrons is larger; this can be measured e.g. in paramagnetic salts at low T (Schottky anomaly: bump in C vs T at $T \rightarrow 0$)

SUMMARY

- $p = T \frac{\partial S}{\partial V}$; with this definition of pressure, moveable partition between systems does not move ($dV=0$) when $p_1 = p_2$
- $dE = TdS - pdV$; first law of thermodynamics
- Two contributions to energy variation: internal energy of the system (“heat”) + work done on the system



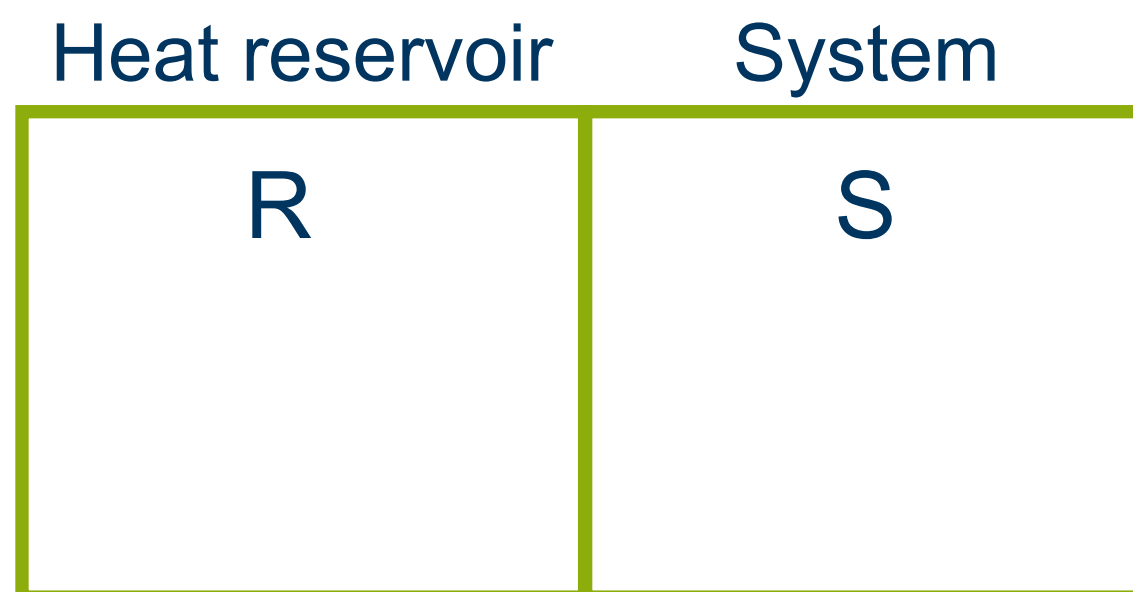
SUMMARY

- Canonical ensemble: we fix the temperature ('reservoir'), the energy can change

- Probability distribution: $p(n) = \frac{e^{-E_n/k_B T}}{\sum_m e^{-E_m/k_B T}}$ (Boltzmann distribution)

- Partition function: $Z = \sum_n e^{-\beta E_n}$

- Independent systems: $Z = \prod_i Z_i$



$$E_R \gg E_S$$
$$T_R = T_S = T$$

Q: can you summarise the properties of the microcanonical ensemble? What we keep fixed, what can change, $p(n)$...?

CONTENTS

- Partition function and what we can do with it
- The thermodynamic limit
- Two state system, studied with the partition function
- Entropy in the canonical ensemble: Gibbs entropy formula
- Entropy in terms of the partition function

PARTITION FUNCTION AND AVERAGE ENERGY

Some calculations:

$$\frac{\partial}{\partial \beta} \log Z = \frac{1}{Z} \frac{\partial Z}{\partial \beta} = \sum_n \frac{1}{Z} (-E_n) e^{-\beta E_n}$$

Average energy of the system:

$$\langle E \rangle = \sum_n p(n) E_n = \sum_n \frac{E_n e^{-\beta E_n}}{Z}$$

Partition function:

$$Z = \sum_n e^{-\beta E_n}$$

Average energy as a function of the partition function:

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \log Z$$

← If we know the partition function of a system, we can derive its average energy

PARTITION FUNCTION AND ENERGY FLUCTUATIONS

Partition function:

$$Z = \sum_n e^{-\beta E_n}$$

Some calculations:

$$\frac{\partial^2}{\partial \beta^2} \log Z = \frac{\partial}{\partial \beta} \left(\frac{1}{Z} \frac{\partial Z}{\partial \beta} \right) = -\frac{1}{Z^2} \frac{\partial Z}{\partial \beta} \frac{\partial Z}{\partial \beta} + \frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} = -\left(\frac{1}{Z} \frac{\partial Z}{\partial \beta} \right)^2 + \sum_n \frac{1}{Z} (-E_n)^2 e^{-\beta E_n} = -(-\langle E \rangle)^2 + \langle E^2 \rangle$$

$$\frac{\partial}{\partial \beta} \log Z = \frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\langle E \rangle$$

$$\langle E^2 \rangle = \sum_n p(n) E_n^2 = \sum_n \frac{e^{-\beta E_n}}{Z} E_n^2$$

Spread of energy about the mean:

$$\begin{aligned} \Delta E^2 &= \langle (E - \langle E \rangle)^2 \rangle = \langle E^2 + \langle E \rangle^2 - 2E \langle E \rangle \rangle = \\ &= \langle E^2 \rangle + \langle E \rangle^2 - 2\langle E \rangle \langle E \rangle = \langle E^2 \rangle - \langle E \rangle^2 \end{aligned}$$

Variance

Hence:

$$\Delta E^2 = \frac{\partial^2}{\partial \beta^2} \log Z = -\frac{\partial \langle E \rangle}{\partial \beta}$$

We can express also the energy fluctuations as a function of the partition function

ENERGY FLUCTUATIONS AND HEAT CAPACITY

From before:

$$\Delta E^2 = \frac{\partial^2}{\partial \beta^2} \log Z = -\frac{\partial \langle E \rangle}{\partial \beta}$$

$$\beta := \frac{1}{k_B T}$$

Heat capacity in canonical ensemble:

$$C_V = \left. \frac{\partial \langle E \rangle}{\partial T} \right|_V$$

In microcanonical, we had fixed energy E ; now, average energy $\langle E \rangle$

$$C_V = \left. \frac{\partial \langle E \rangle}{\partial T} \right|_V = \frac{\partial \langle E \rangle}{\partial \beta} \frac{\partial \beta}{\partial T} \bigg|_V = \frac{\partial \langle E \rangle}{\partial \beta} \left(-\frac{1}{k_B T^2} \right) = -\Delta E^2 \left(-\frac{1}{k_B T^2} \right)$$

Probabilistic fluctuations in the energy of the system

$$\Delta E^2 = k_B T^2 C_V$$

Capability of the system to absorb energy

- If C_V is large, the system can take a lot of energy w/o raising T too much
- The energy fluctuations in the systems are related to the ability of systems of dissipate or absorb energy → first formulation of the **fluctuation-dissipation theorem**

ENERGY FLUCTUATIONS AND THE THERMODYNAMIC LIMIT

Relative amplitude of fluctuations: $\frac{\Delta E}{E}$

Scale of energy fluctuations: $\Delta E^2 \propto C_V \propto N \rightarrow \Delta E \propto \sqrt{N}$

Scale of energy: $E \propto N$

Relative amplitude of fluctuations: $\frac{\Delta E}{E} \propto \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}}$

Thermodynamic limit: $N \rightarrow \infty$ (“lots of particles”)

With $N \rightarrow \infty$, $\Delta E \rightarrow 0$: the energy peaks closer and closer to $\langle E \rangle$, essentially fixed

→ **In the thermodynamic limit, the microcanonical and the canonical ensembles coincide**

BACK TO THE TWO STATE SYSTEM/ 1

As before: N **non-interacting** particles, either spin up $|\uparrow\rangle$ or spin down $|\downarrow\rangle$

We try to recover previous results, using the partition function

Energy of the states (by assumption):

$$E_{\downarrow} = 0 \quad E_{\uparrow} = \epsilon$$

Partition function for a single particle:

$$Z_1 = \sum_n e^{-\beta E_n} = e^0 + e^{-\beta\epsilon} = 1 + e^{-\beta\epsilon} = 2e^{-\beta\epsilon/2} \cosh(\beta\epsilon/2)$$

$\cosh x = \frac{1 + e^{-2x}}{2e^{-x}}$

Q: what are the 'n' are?

The N particles are independent systems:

Q: what is the partition function of a system composed of a number of **independent** sub-systems?

$$Z = 2^N e^{-N\beta\epsilon/2} \cosh^N(\beta\epsilon/2)$$

With the partition function, we can calculate average energy, energy fluctuations, heat capacity ...

BACK TO THE TWO STATE SYSTEM/ 2

$$Z = 2^N e^{-N\beta\epsilon/2} \cosh^N(\beta\epsilon/2)$$

$$\begin{aligned} \langle E \rangle &= -\frac{\partial}{\partial\beta} \log Z = \\ &= -\frac{2^{-N} e^{N\beta\epsilon/2}}{\cosh^N(\beta\epsilon/2)} \left(2^N (-N\epsilon/2) e^{-N\beta\epsilon/2} \cosh^N(\beta\epsilon/2) + 2^N e^{-N\beta\epsilon/2} N \cosh^{N-1}(\beta\epsilon/2) \sinh(\beta\epsilon/2) \epsilon/2 \right) = \\ &= \frac{N\epsilon}{2} \left(1 - \frac{\sinh(\beta\epsilon/2)}{\cosh(\beta\epsilon/2)} \right) = \frac{N\epsilon}{2} (1 - \tanh(\beta\epsilon/2)) \end{aligned}$$

Some algebra:

$$\langle E \rangle = \frac{N\epsilon}{2} (1 - \tanh(\beta\epsilon/2)) = \frac{N\epsilon}{2} \left(1 - \frac{e^{\beta\epsilon} - 1}{e^{\beta\epsilon} + 1} \right) = \frac{N\epsilon}{2} \frac{2}{e^{\beta\epsilon} + 1}$$

Same result as from the microcanonical ensemble:

$$\frac{E}{N\epsilon} = \frac{1}{e^{\epsilon/k_B T} + 1} = \frac{N_{\uparrow}}{N}$$



usually, using Z is easier than counting states

Q: what do you remember about entropy?

WHAT WE ALREADY KNOW ABOUT ENTROPY

Number of microstates

$$S(E) = k_B \log \Omega(E)$$

In the microcanonical ensemble, all these states have fixed energy

Boltzmann's constant: $k_B \sim 1.381 \times 10^{-23} \text{ J K}^{-1}$

Entropy adds in **isolated** systems

Non-isolated systems: $S \geq \sum_i S_i$, second law of thermodynamics

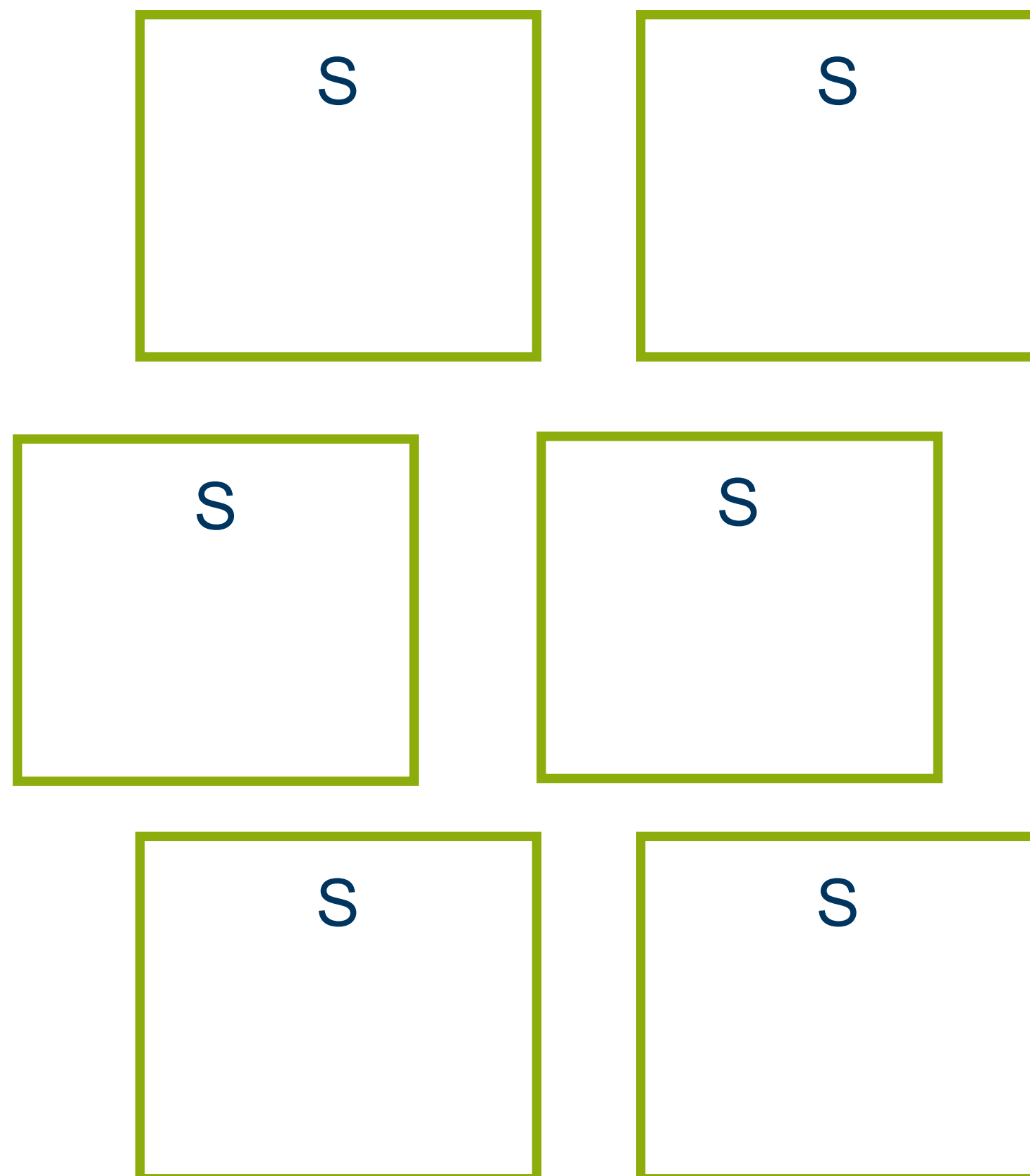
We can calculate temperature from entropy: $\frac{1}{T} = \frac{\partial S}{\partial E}$

First law of thermodynamics: $dE = TdS - pdV$

now, we want a definition of entropy that works in the canonical state

ENTROPY IN THE CANONICAL ENSEMBLE/ 1

We want a definition of entropy in the canonical ensemble; we will build on what we know about entropy in the microcanonical ensemble



We create $W \gg 1$ identical copies of the canonical system S ; each copy is in one of the microstates available to S

Probability of finding one of the copy of S in the microstate $|n\rangle : p(n)$

With $W \gg 1$ copies of the system, $p(n)W$ systems are $|n\rangle$

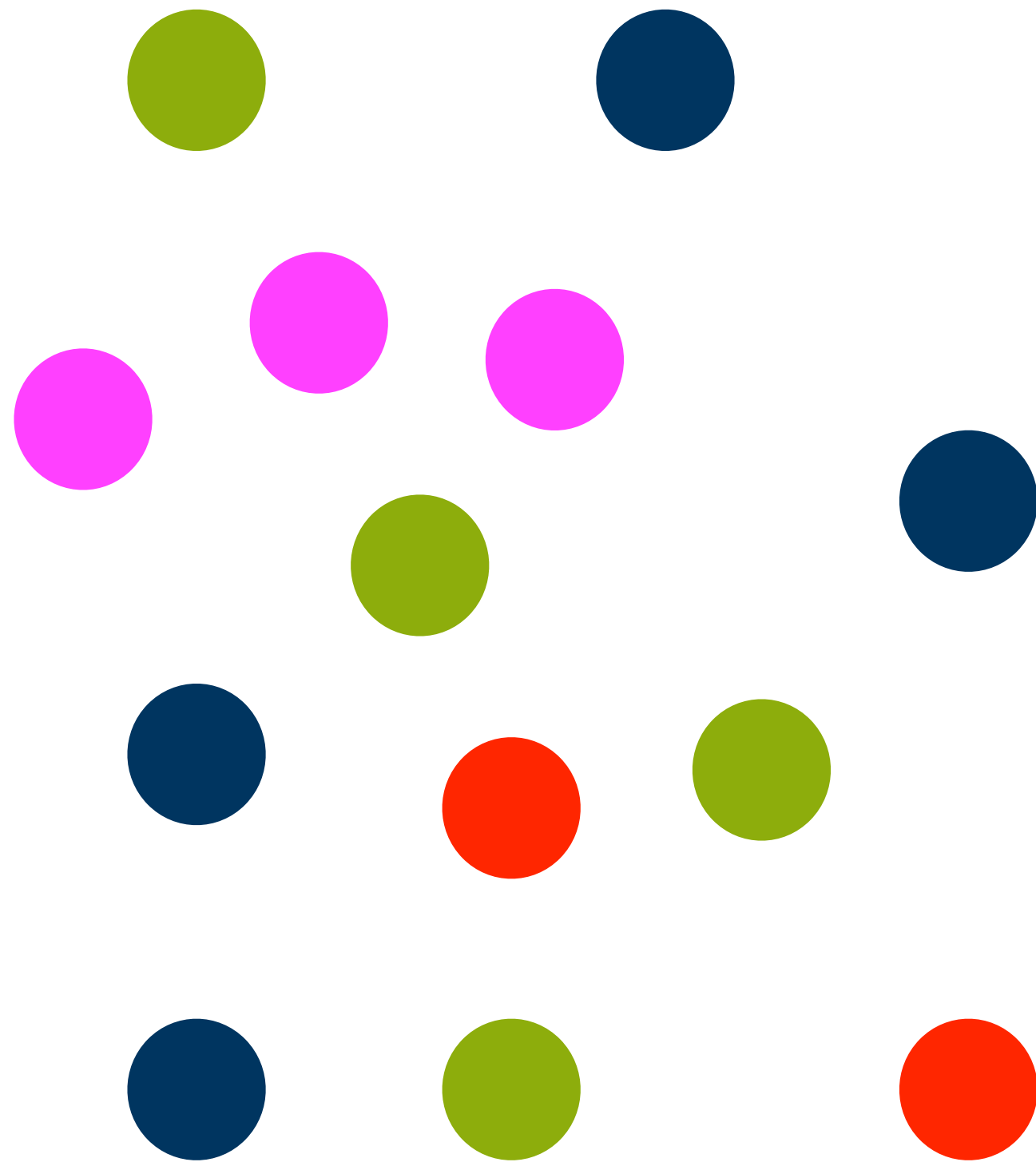
With $W \gg 1$ copies of S

- we have translated **probabilities into eventualities**
- Now we have a number of available microstates: we can think as we did in microcanonical ensembles $\rightarrow$ we can use the definition of S in microcanonical ensembles

$$S = k_B \log \Omega$$

We need the number of microstates Ω

PARTITIONS/ 1



n objects

k groups

Assign n_1 objects to group g_1 , n_2 objects to group g_2 , ..., with $\sum_{i=1}^k n_i = n$

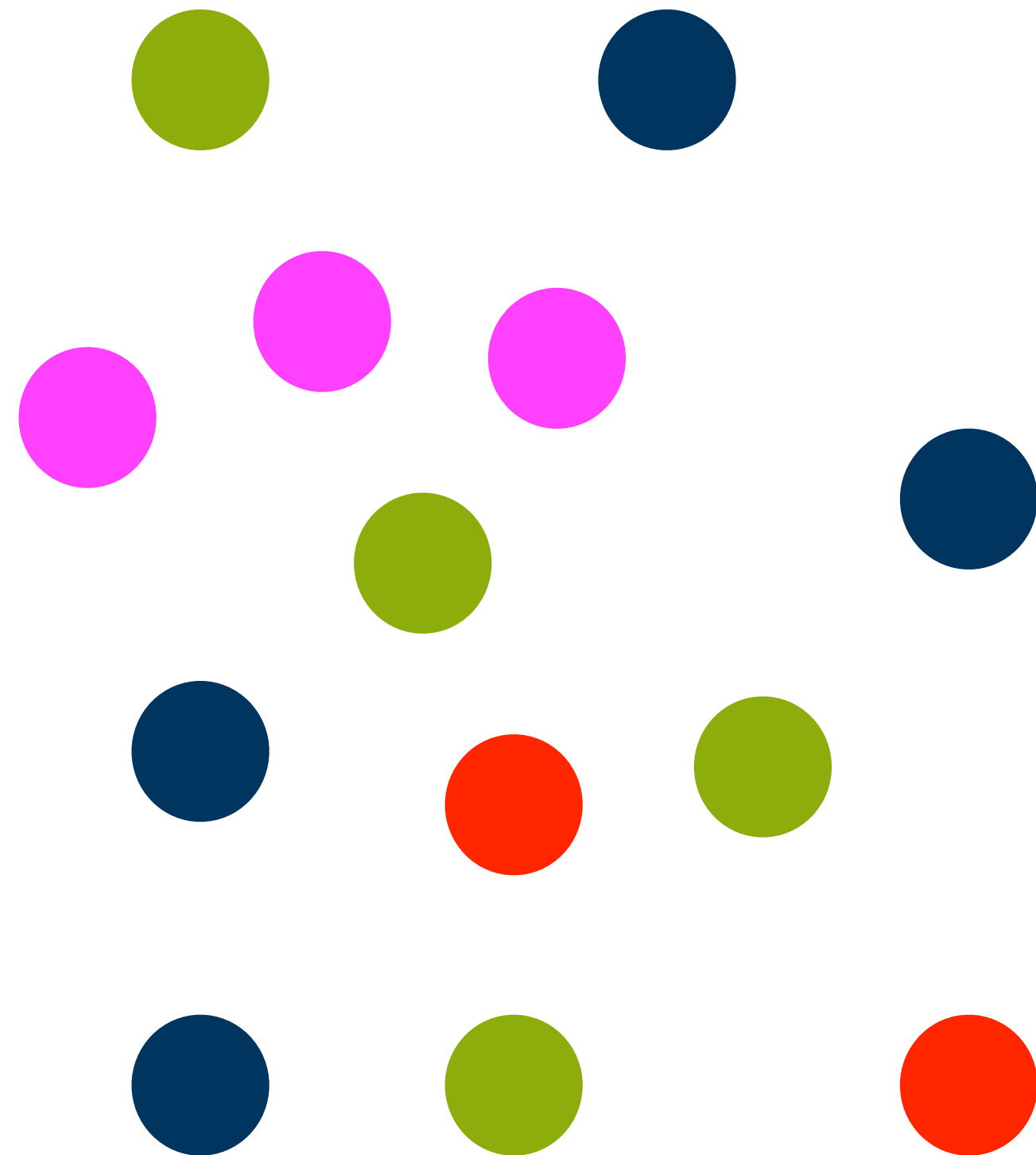
Each object is assigned to one group, order does not matter

Total number of possible partitions of n object in k groups, with $n_1, n_2, \dots, n_k$ known:

$$P_{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$$

PARTITIONS/ 2

How is this useful for us?



n , total number of elements $\rightarrow W$, the number of copies

k , number of partitions $\rightarrow n$, the number of states

$n_1, n_2, \dots, n_k$, number of elements per partition $\rightarrow p(1)W, p(2)W, \dots$ number of systems in state $|1\rangle, |2\rangle, \dots$

$$P_{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$$



$$\Omega = \frac{W!}{(p(1)W)!(p(2)W)! \dots} = \frac{W!}{\prod_n (p(n)W)!}$$

Number of microstates, now we can calculate the entropy

ENTROPY IN THE CANONICAL ENSEMBLE/ 2

$$S = k_B \log \Omega \quad \longrightarrow \quad S = k_B \log \left(\frac{W!}{\prod_n (p(n)W)!} \right)$$

Stirling: $\log N! = N \log N - N + \cancel{1}$ Negligible w.r.t. other terms

$$\begin{aligned} \log \left(\frac{W!}{\prod_n (p(n)W)!} \right) &= \log(W!) - \sum_n \log((pW)!) = W \log W - \cancel{W} + \cancel{1} - \left(\sum_n (pW \log(pW) - \cancel{pW} + \cancel{1}) \right) = \\ &= \cancel{W \log W} - W \sum_n p \log p - \cancel{W \log W \sum p} = -W \sum_n p \log p \end{aligned}$$

$$S_k = k_B \log \Omega = -k_B W \sum_n p(n) \log p(n) \quad \longleftarrow \quad \text{Entropy for all } W \text{ copies of the system.}$$

Q: if this is the entropy of the W copies of the system S , can you calculate the entropy of the system S ? Using which property of entropy?

ENTROPY AS A FUNCTION OF THE PARTITION FUNCTION/ 1

$$S = -k_B \sum_n p(n) \log p(n)$$

Entropy for the original system S

Gibbs entropy formula

Q: does the - sign surprises you?

Q: try applying Gibbs formula to the microcanonical ensemble: what do you get? (see next page)

Probability distribution in canonical ensemble: Boltzmann distribution

$$p(n) = \frac{e^{-E_n/k_B T}}{\sum_m e^{-E_m/k_B T}}$$

$$\sum_n \frac{e^{-\beta E_n}}{Z} = \sum_n p(n) = 1$$

$$\begin{aligned} S &= -k_B \sum_n p \log p = -\frac{k_B}{Z} \sum_n e^{-\beta E_n} \log \left(\frac{e^{-\beta E_n}}{Z} \right) = -\frac{k_B}{Z} \sum_n e^{-\beta E_n} (-\beta E_n - \log Z) = \\ &= +\frac{k_B \beta}{Z} \sum_n e^{-\beta E_n} E_n + k_B \log Z \sum_n \frac{e^{-\beta E_n}}{Z} = \frac{k_B \beta}{Z} \sum_n e^{-\beta E_n} E_n + k_B \log Z \end{aligned}$$

GIBBS FORMULA APPLIED TO MICROCANONICAL ENSEMBLE

$$S = -k_B \sum_n p(n) \log p(n)$$

Microcanonical ensemble: $p(n) = \frac{1}{\Omega}$

$$S = -k_B \sum_n p(n) \log p(n) = -k_B \sum_n \frac{1}{\Omega} \log(1/\Omega) = -k_B \frac{1}{\Omega} \sum_n (\log 1 - \log \Omega) = +k_B \frac{1}{\Omega} \log \Omega \sum_n 1 = k_B \log \Omega$$

0

Number of microstates in state $|n\rangle$, i.e. with energy $E = E_n$

This is the microcanonical ensemble: all the microstates have energy $E = E_n$

$$\sum_n 1 = \Omega$$

ENTROPY AS A FUNCTION OF THE PARTITION FUNCTION/ 2

From before:

$$S = \frac{k_B}{Z} \sum_n e^{-\beta E_n} E_n + k_B \log Z \quad Z = \sum_n e^{-\beta E_n} \quad \beta = \frac{1}{k_B T}$$

Let's try:

$$\frac{\partial}{\partial T} (T \log Z) = \log Z + T \frac{1}{Z} \frac{\partial Z}{\partial T} = \log Z + \frac{T}{Z} \sum_n e^{-E_n/k_B T} \left(-\frac{1}{T^2} \right) \left(-\frac{E_n}{k_B} \right) =$$
$$\log Z + \frac{1}{k_B T} \frac{1}{Z} \sum_n E_n e^{-E_n/k_B T}$$

$$S = k_B \frac{\partial}{\partial T} (T \log Z)$$

Now we have a formula for the entropy as a function of the partition function Z

MICROCANONICAL VS CANONICAL ENSEMBLES

Microcanonical vs canonical ensembles: different probability distributions so in principle different entropies

But in the thermodynamical limit, $N \rightarrow \infty$, entropy (as all observables) must coincide with the two distribution; let's try

Canonical ensemble:

Z written as a sum over **states** n :

$$Z = \sum_n e^{-\beta E_n}$$

Z written as a sum over **energy levels** E_i :

$$Z = \sum_{\{E_i\}} \Omega(E_i) e^{-\beta E_i}$$

Degeneracy factor: how many microstates we have at the same energy level E_i

Sum of exponentials with extremely large arguments, since $N \gg 1$, $\sum_{\{E_i\}}$ is dominated by its larger term

$$\frac{\partial}{\partial E} (\Omega(E) e^{-\beta E})_{E=E_\star} = 0$$

$E_\star$ is not the maximum energy, but the energy that maximizes $\Omega(E) e^{-\beta E}$

MICROCANONICAL VS CANONICAL ENSEMBLES

$$Z = \sum_{\{E_i\}} \Omega(E_i) e^{-\beta E_i} \xrightarrow{N \gg 1} Z \sim \Omega(E_\star) e^{-\beta E_\star} \quad \leftarrow \text{(dominant term in the } \sum_{\{E_i\}} \text{)}$$

Now, let's calculate $\langle E \rangle$ and S with this approximated partition function

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \log Z = -\frac{\partial}{\partial \beta} \log (\Omega(E_\star) \exp^{-\beta E_\star}) = -\frac{1}{\Omega(E_\star) \exp^{-\beta E_\star}} \Omega(E_\star) \exp^{-\beta E_\star} (-E_\star) = E_\star$$

$$k_B \frac{T}{Z} \frac{\partial Z}{\partial T} = k_B \frac{T}{Z} \Omega(E_\star) \frac{\partial}{\partial T} e^{-E_\star/k_B T} = k_B \frac{T}{Z} \Omega(E_\star) (-E_\star/k_B) \left(-\frac{1}{T^2}\right) e^{-E_\star/k_B T} = \frac{E_\star}{T} \frac{\Omega(E_\star) e^{-E_\star/k_B T}}{Z} = \frac{E_\star}{T}$$

$$S = k_B \frac{\partial}{\partial T} (T \log Z) = k_B \log Z + k_B \frac{T}{Z} \frac{\partial Z}{\partial T} = k_B \log \Omega(E_\star) + k_B (-\beta E_\star) + \frac{E_\star}{T} = k_B \log \Omega(E_\star)$$

We see that in the thermodynamic limit (when $N \rightarrow \infty$, which allows us to use $Z \sim \Omega(E_\star) e^{-\beta E_\star}$) physical observables (e.g.: E , S) coincide in the microcanonical and canonical ensemble

SUMMARY

- The partition function tells us a lot on a system:

- Average energy: $\langle E \rangle = -\frac{\partial}{\partial \beta} \log Z$

- Energy fluctuations: $\Delta E^2 = \frac{\partial^2}{\partial \beta^2} \log Z$

- We revisited the two state system using Z : a bit easier to calculate $\langle E \rangle$

- Entropy in the canonical ensemble: Gibbs entropy formula, $S = -k_B \sum_n p(n) \log p(n)$ (works also for microcanonical ensemble)

- Entropy in the canonical ensemble in terms of the partition function: $S = k_B \frac{\partial}{\partial T} (T \log Z)$

- Microcanonical vs canonical ensembles:

- In the thermodynamic limit, $N \rightarrow \infty$, $S_c \rightarrow S_{\mu c}$, with $E = \langle E \rangle$

References: Tong, Sec. 1.3.2 to 1.3.3